

Question 1

- O (Lớp gốc / object)
- class A(O) — foo
- class B(O)
- class C(O) — foo
- class D(A, B)
- class E(C, A)
- class F(D, B, E)

a) Apply the Method Resolution Order (MRO) algorithm of Python 3 to find the class search path of each class in the given class schema. If there are errors in any class search path (inconsistent hierarchy/MRO conflict), modify the order of superclasses in the class definition(s) to make all class search paths successful.

b) Given x is an object of class F, which method foo() will be called by x.foo() using your successful class schema?

Question 2

Define class Exp and its subclasses (BinExp, UnExp, IntLit, FloatLit, and Var if applicable for variables) in Python.

Write the method printPrefix() for these classes to return/print the expression in prefix format.

Create an object representing the arithmetic expression $(x+0.2) \times 3$, assign it to variable t, and print the prefix form of the expression represented by t.

Question 3

Based on the classes defined in Question 2, write the method `eval()` that evaluates an expression and returns a `Number` object.

The operators for binary expressions are `+`, `-`, `*`, `/`, and for unary expressions are `+`, `-`.

Assume that the value of all variables (e.g., `x`) is 1.

For example, `print(t.eval().num)` should output 3.6.